

StaND4LQ

State of the Art & New Directions for
Lattice Gauge Theory and Quantum Simulation

Pisa · 30 September – 2 October 2026

Roberto Dionisio

Andrea Giorgieri

Marco Intini

Margherita Sagina

University of Pisa · organized by PhD students, supported by INFN Sezione di Pisa



Benvenuti a Pisa and welcome to the **first edition** of StaND4LQ.

This workshop is organised **entirely by PhD students** of the University of Pisa.

No senior committee and no registration fee.

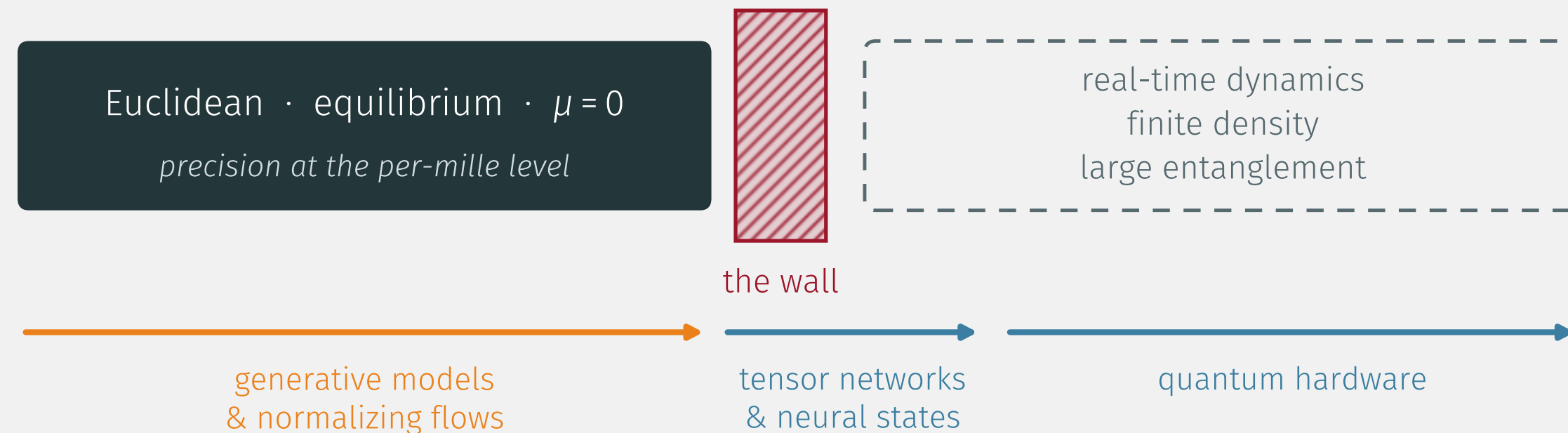
Three days, **eighteen talks**, **nine invited speakers**, and one long discussion with no talks at all.

Lattice field theory is now a precision science

...and precisely *because* it is, we can see clearly where it stops.

STATE OF THE ART

NEW DIRECTIONS



The wall is not a technicality. It is a **sign problem**, an ill-posed **analytic continuation**, and a Hilbert space that grows **exponentially**.

What blocks us

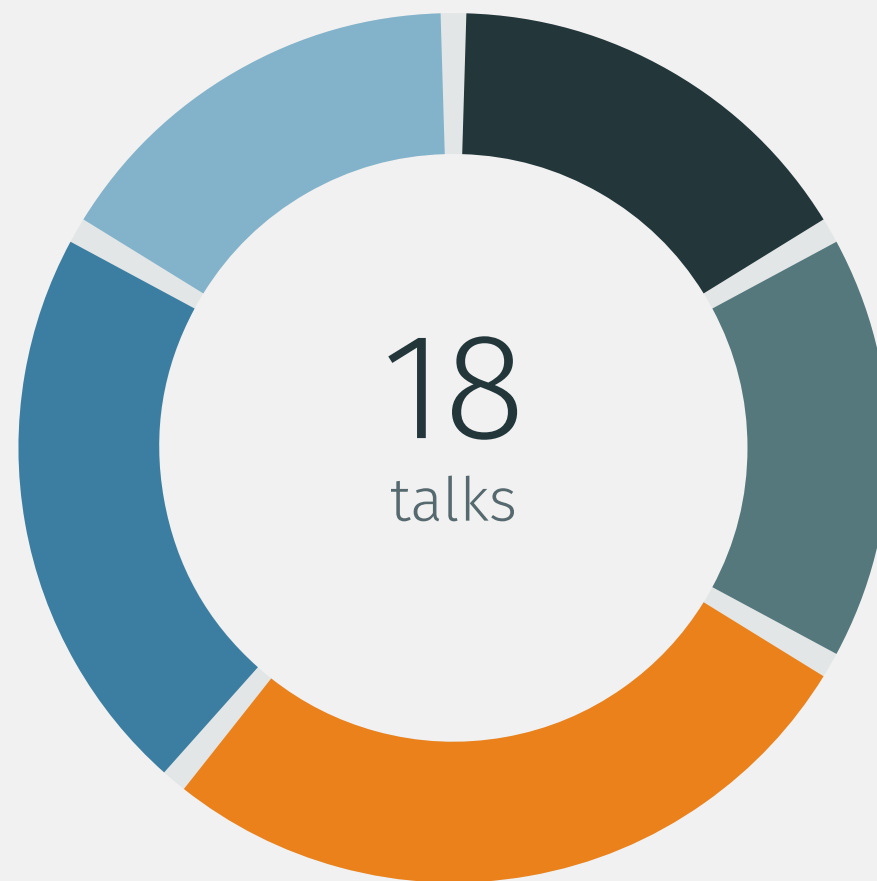
- efficient **Monte Carlo** sampling
- **real-time** dynamics
- systems at **finite density**
- gauge theories on **quantum devices**

Who is needed

- lattice **field theory**
- **quantum information**
- **machine learning**
- quantum **many-body** physics

None of these four problems belongs to a single community
and that is the entire premise of this meeting.

Eighteen talks over three days.



- Precision QCD & flavour
3 talks · 17%
- Topology & finite- T QCD
3 talks · 17%
- Generative models & sampling
5 talks · 28%
- Quantum simulation of gauge theories
4 talks · 22%
- Quantum many-body & control
3 talks · 17%

Nine invited speakers

Claudio Bonanno

University of Bern

Bipasha Chakraborty

University of Southampton

Giuseppe Gagliardi

INFN, Sezione di Roma Tre

Giulia Piccitto

University of Catania

Matteo Wauters

University of Trento

Pietro Butti

University of Southern Denmark

Pierpaolo Fontana

Universitat Autònoma de Barcelona

Alessandro Nada

Università di Torino · INFN Torino

Simone Romiti

University of Bern

Nine contributed talks from early-career researchers sit alongside them.

A first edition, in numbers

31

registered participants

18

talks in three days

5

countries represented

0

registration fee

Italy · United Kingdom · Switzerland · Denmark · Spain

For a first edition, assembled by four PhD students, we are quite happy!

Thursday 1 October · 12:10 – 13:30

An hour and twenty minutes with **no speaker and no slides**.

It is the only part of the programme we cannot prepare for you.

If you came here to find a collaborator rather than to give a talk,
this is the slot that matters.

Venue

Polo Fibonacci, Building A
Aula Magna, Dept. of Mathematics
Largo Bruno Pontecorvo 3, Pisa

Coffee breaks are offered by the organisation, just outside this room.

Talk slots

Invited 50 minutes, contributed 30,
questions included.

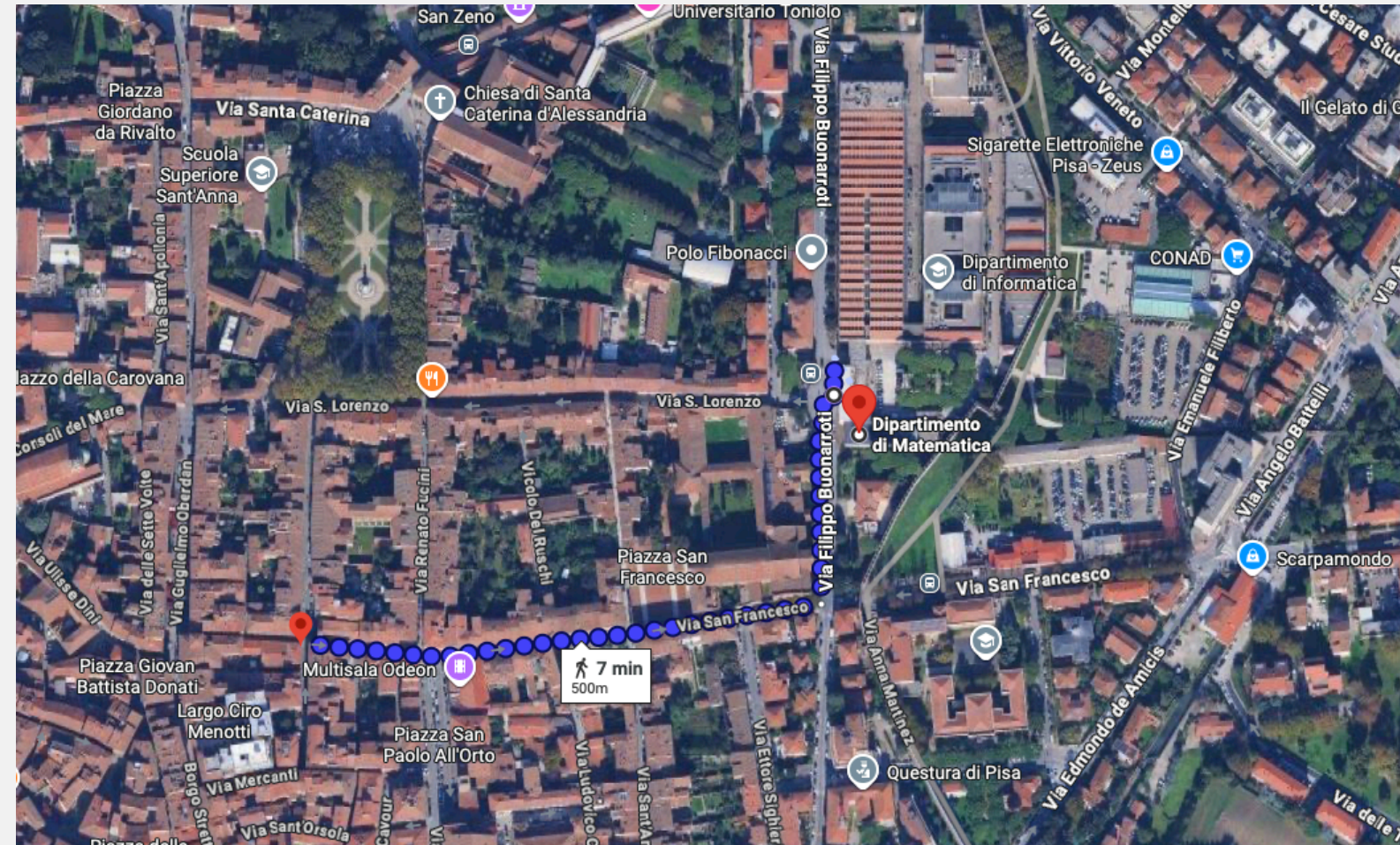
Anything at all: find any of the four of us, or write to
stand4lq_workshop@lists.infn.it

Ristorante **La Grotta**

Thursday 1 October

20:00

7 minutes on foot from this room,
straight down Via San Francesco.





This workshop is funded by the **University of Pisa** and **INFN, Sezione di Pisa**, which is why nobody paid to walk through the door.

Coffee for three days, and a first edition that did not have to charge for itself.

To our **nine invited speakers**, for agreeing to come to a workshop that did not yet exist.

To everyone who **submitted an abstract**, and to the nine early-career researchers giving contributed talks.

One request · the people in this room work on Euclidean Monte Carlo, on tensor networks, and on quantum hardware. Sit next to someone whose methods you do not use.

Enjoy the workshop!